Assignment 2

~~Assigment~~ 2.

```
int n=5
for(int i=1;i<=n;i++){
 for(int j=1;j<=n;j++){
   if(i==1||j==1||i==n||j==n){
        sop("*");
   }
   else{
        sop(" ");
   }
 }
 sop()  sopln();
}
```

(1,1)	(1,2)	(1,3)	(1,4)	(1,5)
(2,1)	(2,2)	(2,3)	(2,4)	(2,5)
(3,1)	(3,2)	(3,3)	(3,4)	(3,5)
(4,1)	(4,2)	(4,3)	(4,4)	(4,5)
(5,1)	(5,2)	(5,3)	(5,4)	(5,5)

Dry run

outerloop i=1 i<=5 i++

↓

innerloop j=1 j<=5 j++

↓

if(i==1||j==1||i==n||j==n){

sop("*");

}

j=2 2<=5 j++

↓

j=1 *

j=3 3<=5 j++

j=1 *

j=4 4<=5 j++

j=1 *

j=5 4<=5 j++

*

j=6 6<=5 ✗

i=2 2<=5 i++

↓

j=1 same process

if condition check

j=1 *

j=2 *

j=3 * — —

j=4 * — —

j=5 *

i=3 j=1 *

j=2 * — —

j=3 — — —

j=4 — — —

j=5 *

i=4 j=1 *

j=5 *

j=2,3,4 — — —

i=5 j=1 *

j=5 *

j=5 *

j=5 *

j=5 *

Q. int n = 5
for(int i = 1; i<=n; i++){
for(int j = 1; j<=n; j++){
if(i==j || i+j==n+1){
sop(*);
}
else{
sop(---);
}
}sopln();
}

n=3

(1,1) (1,2) (1,3)

(2,1) (2,2) (2,3)

(3,1) (3,2) (3,3)

i==j किंवा i+j = n+1

By run

outer i=1; i<=5 i++

inner j=1 i<=5 j++

if(i==j){

sop → *

}

j=2 2<=5 j++

× if (i==j × || i+j=5+1

2+2 = 5+1

3 = 6

else {

}

}

j=3 3<=5 j++

× if

else {

sop → --

}

j=4 4<=5 j++

else {

--

}

j=5 5<=5 j++ → * i+j=

j=6 6<=5 ×

i=2 2<=5 i++

j की Process same

i==j, i+j = n+1 check करायचा

j=1 --

j=2 *

j=3 --

j=4 * i+j = 2+4 = 6

j=5 --

i=3 3<=5

j=1 --

j=2 --

j=3 *

j=4 --

j=5 --

i=4 4<=5

j=1 --

j=2 * i+j=6

j=3 --

j=4 i==j * =8

j=5 ---

i=5 5<=5

j=1 i+j=6 = *

j=2 --

j=3 --

j=4 --

j=5 i==j = *

i=6 6<=5 ×

innerloop संपला.

Q.

```
int n=5
for(int i=1; i<=n; i++){
for(int j=1; j<=i; j++){
    sop("*");
}
sopln();
}
```

Pattern	j=1	j=1	j=1	j=1	j=1
i=1	*				
i=2	*	*			
i=3	*	*	*		
i=4	*	*	*	*	
i=5	*	*	*	*	*

i=1 1<=5 i++

j=1 1<=i(1) j++

*

j=2 2<=1 ✗

i=2 2<=5 i++

j=1 1<=i(2) j++

*

j=2 2<=2 j++

*

j=3 3<=2 ✗

i=3

j=1 *

j=2 *

j=3 *

j=4 ✗

i=4

j=1 *

j=2 *

j=3 *

j=4 *

j=5 ✗

i=5

j=1 *

j=2 *

j=3 *

j=4 *

j=5 *

j=5 ✗

i=6 6<=5 ✗

*

* *

* * *

* * * *

* * * * *

Q.

```
int n=5
for(int i=1; i<=n; i++){
```

```
int n=5
for(int i=n; i>=1; i--){
for(int j=1; j<=i; j++){
    sop("*");
}
sopln();
}
```

5 → * * * * *
4 → * * * *
3 → * * *
2 → * *
1 → *

j=1 1<=5 ↓ *
j=2 *
j=3 *
j=4 *
j=5 *
j=6 X

5>=1 ↓ i--

i=4 4>=1 ↓ i--
j=1 1<=4 ↓ *
j=2 *
j=3 *
j=4 *
j=5 X

i=3. 3>=1 ↓
j=1 *
j=2 *
j=3 *
j=4 X → i--

i=2
j=1=*
j=2=*
j=3 X

j=1 1>=1 ↓ 0>=1 X
j=1 *
j=2 X

Q. int n=4
for(int i=1; i<=n; i++){
for(int j=1; j<=n; j++){
if((i+j)%2==0){
sop("*");
}
else{
sop(#);
}
}
sopln();
}
}

*	#	*	#
#	*	#	*
*	#	*	#
#	*	#	*

(1,1)	(1,2)	(1,3)	(1,4)
*		*	
(2,1)	(2,2)	(2,3)	(2,4)
*	*		*
(3,1)	(3,2)	(3,3)	(3,4)
*		*	
(4,1)	(4,2)	(4,3)	(4,4)
	*		*

i=1 1<=4 j++
↓
j=1 1<=4
↓
i+j%2=0 ✓
*
j=2 → i+j%2=0 X
j=3 → i+j%2=0 ✓
j=4 → i+j%2=0 X
j=5 X

i=2 2<=4 j++
↓
j=1 1<=4
i+j%2=0 X
j=2 2+2=0 ✓
*
j=3 2+3=0 X
j=4 2+4=0 ✓

i=3
j=1 3+1=0 ✓ *
j=2 3+2!=0 X #
j=3 3+3=0 ✓ *
j=4 3+4!=0 X #
j=5 X

i=4
j=1 4+1!=0 X #
j=2 4+2=0 ✓ *
j=3 4+3%2!=0 X #
j=4 4+4%2=0 ✓ *